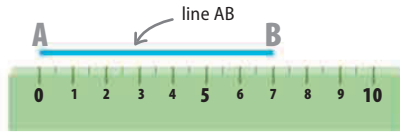


Using a ruler

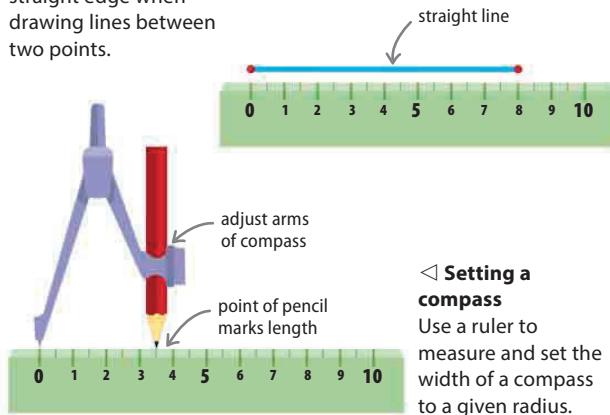
A ruler can be used to measure straight lines and the distances between any two points. A ruler is also necessary for setting the arms of a compass to a given distance.



◁ **Measuring lines**
Use a ruler to measure straight lines or the distance between any two given points.

▷ Drawing lines

A ruler is also used as a straight edge when drawing lines between two points.



◁ **Setting a compass**
Use a ruler to measure and set the width of a compass to a given radius.

Other tools

Other tools may prove useful when creating drawings and diagrams in geometry.



△ Set square

A set square looks like a right triangle and is used for drawing parallel lines. There are two types of set square, one has interior angles 90°, 40° and 45°, the other 90°, 60°, and 30°.

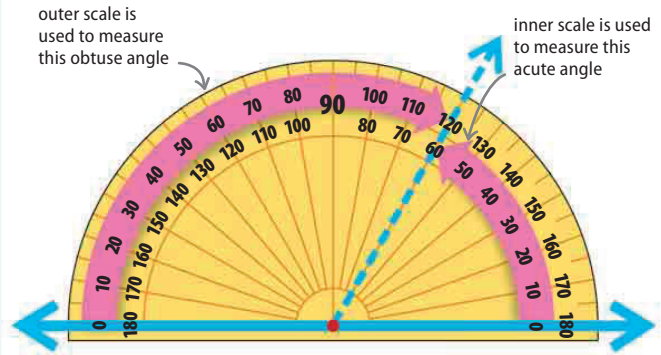


△ Calculator

A calculator provides a number of key options for geometry calculations. For example, functions such as Sine can be used to work out the unknown angles of a triangle.

Using a protractor

A protractor is used to measure and draw angles. It is usually made of transparent plastic, which makes it easier to place the center of the protractor over the point of the angle. When measuring an angle, always use the scale starting with zero.



▽ Measuring angles

Use a protractor to measure any angle formed by two lines that meet at a point.



Extend the lines if necessary to make reading easier.

▽ Drawing angles

When given the size of an angle, use a protractor to measure and draw the angle accurately.



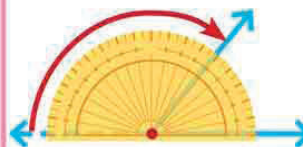
Draw a line and mark a point on it.



Place the protractor over the angle and read the angle measurement, making sure to read up from zero.



Place the protractor on the line with its center over the point. Read the degrees up from zero to mark the point.



The other scale measures the external angle.



Draw a line through the two points, and mark the angle.